MINING DEALER BEST PRACTICE

Identifying Physical Availability Loss Factors through Downtime Stratification by CPI/PN

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

This Best Practice describes the process used to collect, analyze, and report the percent Physical Availability (and other metrics) lost by machine/fleet per CPI number and/or part number.

Currently, top downtime drivers are often reported by machine system. These are either defined by SCMS codes *(https://servicedata.cat.com/Home/SMCSInquiry)* or defined on a site-by-site preference. There is infrequent correlation between the downtime and the part that caused the failure and specifically if there is an existing CPI project that covers that failure mode. This means that both the dealer and Caterpillar know the systems that are causing the top downtime but not the components responsible.

Downtime Stratification is a process that provides granularity to the downtime and can be used to:

- Improve fleet management
- Develop predictive maintenance strategies
- Provide Product Support with information that can assist with CPI ranking and resource allocation

2.0 Best Practice Description

Physical Availably PA is a top tier metric vital to customer production and is a critical element of the customer experience. Downtime is recorded and analyzed to better understand the machine systems responsible for negative impact on PA.

By utilizing the Downtime Stratification approach, a representative from each of the customer, dealer and Caterpillar, from the reliability division and a Caterpillar data analyst can determine PA loss by specific part number and correlate it to appropriate CPI project.

Downtime data is provided to the dealer/Caterpillar by the customer. There are situations where the first step involves the dealer and customer working together to reconcile raw data. Collaboration is required between the customer, dealer, and Caterpillar to evaluate the downtime and associate it with part numbers. Using the 80:20 rule of data distribution as described in *Downtime Priorities, Jack-knife Diagrams, and the Business Cycle by Peter F. Knights*, Appendix A, 80% of the downtime data is analyzed.
Caterpillar and the dealer are then responsible for assigning applicable CPIs to these down events. The Site Performance Manager (SPM) or Technical Representative will analyze the data and calculate top tier metrics per CPI #. The data is then sent to a Stratification Analyst for consolidation. This process is mapped out below in Figures 2.0 & 2.1.

Downtime Stratification Process Map

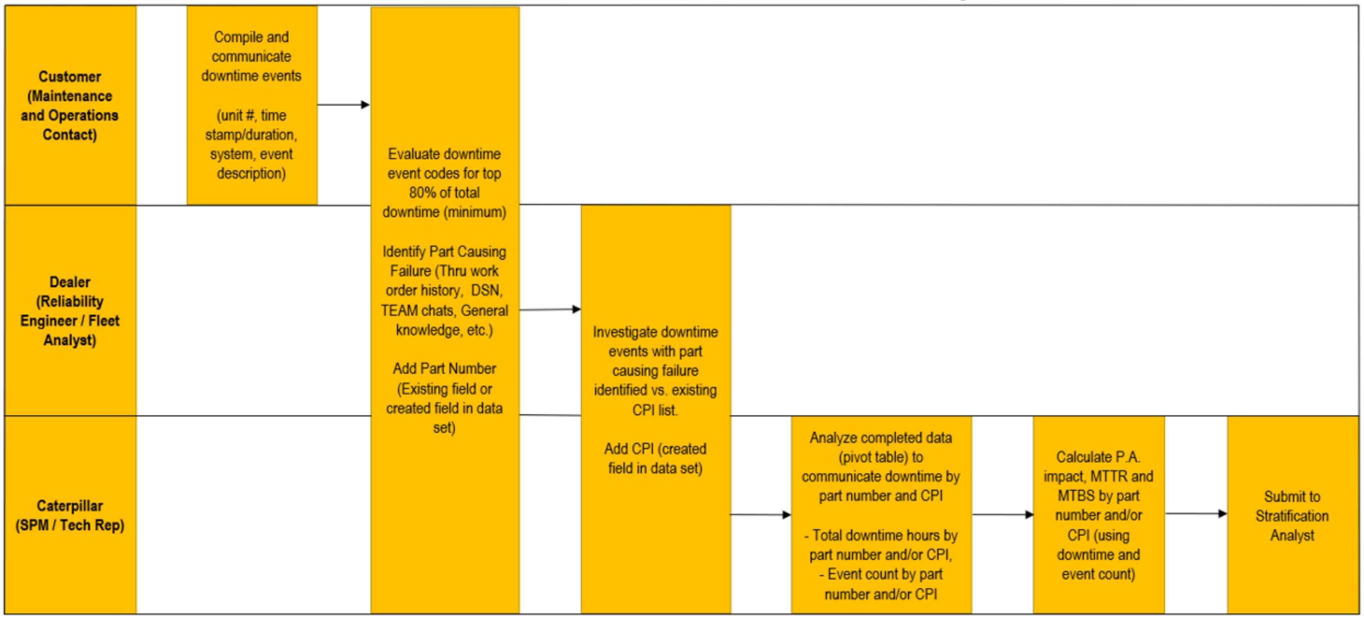

Figure 2.0

Downtime Stratification SIPOC

<table>
<tr><th>Suppliers</th><th>Inputs</th><th>Processes</th><th>Output</th><th>Customers</th></tr>
<tr><td rowspan="3">Mine Maintenance and Operations Contact</td><td>Site downtime data events</td><td rowspan="5">DOWNTIME STRATIFICATION</td><td rowspan="2">Prioritized ICA / PCA</td><td>Mine mainenance and Operations</td></tr>
<tr><td>Repair/Work order information (Parts required to fix down event)</td><td rowspan="2">Dealer</td></tr>
<tr><td>Labor Information (Labor required to fix down event)</td><td rowspan="2">Downtime Impact Metrics (Event count, DT hours, %PA, MTTR, MTBS)</td></tr>
<tr><td>Dealer (Reliability Engineer / Fleet Analyst)</td><td>Part number vs. downtime event(s)</td><td rowspan="2">Caterpilar CPI / Engineering</td></tr>
<tr><td>Caterpillar (SPM / Tech Rep)</td><td>CPI / DPPM number vs. Part Number</td><td>Ranked/Prioritized CPI issues based on Customer Experience</td></tr>
</table>

Figure 2.1

Customers use various systems for capturing machine stoppages. The downtime is reconciled, consolidated, and shared with their dealers and Caterpillar typically through the site's management system.

Downtime information required:

1. Unit ID,
2. Time stamp
3. Downtime Duration
4. Event description

Customer, Caterpillar Site Performance Manager (SPM) or Technical Representative and the Dealer Reliability Engineer (RE) or Fleet Analyst, analyze, at a minimum, the top 80% of downtime and identify the part causing the failure. Where applicable, a corresponding CPI is added.

Resources utilized for this task include:
1. Work order information
2. DSN entries
3. TEAM channels
4. CPI lists
5. General site knowledge of team members

The Caterpillar SPM/Tech Rep utilizes a pivot table to compute, by CPI/PN:
1. Down time hours
2. Physical Availability %
3. Event Court
4. MTTR
5. MTBS

An Excel file, updated monthly by the Stratification Analyst, can be used to analyze the data. See the attached file called *Downtime Stratification Workbook*, Appendix B. The file can range from 3 tabs for a single fleet of machines to an unlimited number of tabs representing an unlimited number of fleets that we want to analyze with combined data.

The Downtime Stratification Workbook should contain separate tabs for:

1. Basic CPI information and top tier metrics associated with each issue (Figure 2.2 and 2.3, Appendix B).

   A report, in Excel format, can be generated using the following website. (Access is limited to Caterpillar Product Support contacts).
   https://grwb.ecorp.cat.com/cpi/Reports/CustomCriteria.cfm
   Under the “Product and Parts” tab, select the model you are interested in from list provided. Then select the items listed in #1 below from the “Outputs” tab. Finally, select “Export Details” to generate the CPI list. This list will aid in assigning specific down event to their respective CPIs and can be filtered for Open, Pending, and Closed Issues as well.

   The first section of the Excel sheet will contain the CPI information.
   - Issue Number
   - List Number
   - Title
   - Primary Status
   - Secondary Status
   - Target Closure Date
   - Risk Level
   - Days Open

Downtime Stratification CPI Issue List

Issue	ID #	List	Title	Primary Status	Scnd Status	Target	Risk Level	Days	ICA	PCA
459805	2	2700	794/6/8 AC Front Wheel Flange Cracking (584-4172)	Open	Active	31-Mar-23	W (Watch)	973	PS47189	
485427	52	542	794 AC, HOUSING GP-RRAX-Gp, AXLE AS-MACHINED, PANHARD ROD MOUNTING, CRACKED (4479247)	Open	Active	31-Dec-23	A (High)	525	M0128925	
478505	29	2700	794/6/8 PARK BRK, SEAL-RECTANGULAR, LEAKS OIL/LUBE, (1750100)	Open	Active	09-Aug-22	W (Watch)	667	PS47087(794/6) M0128576	
479365	4	542	CPI479365 - 794 AC, 796 AC, 798 AC, DRIVETRAIN BLOWER MOTOR FAILURES (303-5861)	Closed	Resolved	01-Nov-22	B (Medium)	595	PS54603	PS47304 M0134645
500158	2004	3360	794 AC GEAR GP-FRONT, PUMP GP-WATER -B (4160612)(20R0896)	Pending	Submitted	31-Dec-22	W (Watch)	236		
385830	46	3360	CPI 385830: C175, 793F, 794, 795F, 797F, Alternator Tensioner Torsion Spring Failures, Guard failures	Closed	Resolved	17-Dec-19	C (Low)	1049	Same Part	
461219	13	542	794 AC, 796 AC, 798 AC Starter Motor Failures (277-7662) [793F Replication Project]	Open	Active	31-Dec-22	A (High)	951	Same Part M0131985	
464779	24	542	796/8 AC HRC Low HP NPI - 5422643 - LEAKS HYDRAULIC FLUID FROM STEERING LINES - 5692102	Open	Active	16-Dec-22	C (Low)	645	TIB M0127729	
438190	100	1132	CPI438190 793F/795F/797F 794/796/798, 6" E-Stat Seizes or Sticks or Non-Responsive (494-2623)	Open	Active	30-Nov-22	B (Medium)	1122	TIB - M0135889	SM - M0117053
439991	78	5279	LMT Alternator and Regulator Failures (437-0125, 421-7193, 424-7245) (Eng)	Closed	Resolved	08-Apr-22	C (Low)	1148		
448166	79	542	794/6/8 LRC Low HP Field Follow - Left & Right Lower Rear Strut Spherical Bearings (431-8494) Cracked	Open	Active	15-Jun-22	B (Medium)	952	Same Part	
475003	25	542	798 AC (also used on 794/6) - UPPER FRONT RIGHT AND LOWER LEFT SUSPENSION CYLINDER PIN FRACTURED AT STEPPED LOCATION IN PIN - 4832755	Open	Active	31-Dec-23	A (High)	695	Same Part PS47349 M0123441	M0126777 M0126783
476875	26	3360	CPI 476875 793F T4 & 794 AC T4, HARNESS GP-WRG-Gp, HARNESS AS-FOAM, OPEN CIRCUIT, (5404514)	Closed	Resolved	29-Jun-22	B (Medium)	685	PS54618 (796) PS47016 (794/8)	
472870	49	542	796/8 AC HRC Low HP NPI - 4422536 - CRACKED BEZEL/DASH PANELS AND MISSING BUTTONS - 4322929	Open	Active	31-Aug-22	B (Medium)	533	Test Iron	
505978	3099	6709	798 AC HRC Low HP NPI - OIL SANDS SPECIFIC - View through right side mirror obstructed by exhaust box (6048945) on OS HP dump body (6020835)	Open	Accepted	31-Jul-22	B (Medium)	111		
431806	910	542	79X LRC Field Follow - Damaged Egress Ladder and Battery Box	Closed	Resolved	17-Oct-19	B (Medium)	212		
451937	54	542	794/6/8 LRC Low HP Field Follow - Cracked Bottom Hydraulic Tank Mounting Strap (442-3480)	Open	Active	30-Nov-22	A (High)	1057	Same Part M0117783	
410103	643	3360	CPI 410103 793F,795F,794F,797F, C175-16 and C175-20 Camshaft Spalling Failures (4883487, 4883489, 4883490, & 4883491)	Closed	Resolved	13-Dec-21	B (Medium)	1491	M0131895	PS46860 M0126425
451924	1190	542	794/6/8 - Front Wheel Mounting Group 447-9174 - Missing Coiled Pin 447-9170	Open	Active	31-May-22	B (Medium)	1071		
485865	1798	4337	All Truck Applications C175 Intake Valve Recession (513-3495) - CSL	Closed	Cancelled	31-Dec-22	W (Watch)	420		
504614	3072	542	794/796/798 AC - AXLE GP-FRONT, AXLE AS-FRONT-MACHINED, CRACKED (443-8883, 447-9166, 539-6288)	Open	Active	31-Dec-22	A (High)	140		
474605	23	542	796/8 AC HRC Low HP NPI - 4466066 KING PIN RUSTED, CORRODED, DAMAGE TO THRUST RINGS AND BEARING	Open	Active	23-Dec-22	B (Medium)	525	Same Part	
498557	1979	2700	796/8 AC HRC Low HP NPI - CRACKED REAR OUTER WHEEL ADAPTERS- 5652968	Closed	Resolved	31-Mar-22	W (Watch)	139		
452506	1204	542	796/8 LRC Low HP Field Follow - Weld Cracked Between Rear Left Frame Lug (442-3153) and Frame Crosstube (536-8505)	Closed	Resolved	17-Sep-20	B (Medium)	158		

Figure 2.2

The tab will also contain formulas for combining data from each fleet (1 tab per fleet) and consolidating it if the project is combining the same model over multiple sites for comparison, as shown in Figure 2.3.

- Downtime Impact (Hours)
- Event Count
- PA loss (%)
- MTBS (Hours) *MTBS is calculated to include **both** scheduled and unscheduled downtime.*
- MTTR (Hours)

Top Tier Metrics

Issue	ID #	Model AC - Site 1	Model AC - Site 2	Model AC - Site 3	AC Fleet - Combined	Model AC - Site 1	Model AC - Site 2	Model AC - Site 3	AC Fleet - Combined	Model AC - Site 1	Model AC - Site 2	Model AC - Site 3	AC Fleet - Combined	Model AC - Site 1	Model AC - Site 2	Model AC - Site 3	AC Fleet - Combined	Model AC - Site 1	Model AC - Site 2	Model AC - Site 3	AC Fleet - Combined
459805	2	2,906	247	-	3,153	22	-	-	22	3.24%	0.47%	-	3.71%	2,153	-	-	2,153	132	-	-	143
485427	52	-	422	-	422	-	8	-	8	-	0.80%	-	0.80%	-	6,517	-	6,517	-	53	-	53
464779	24	33	16	7	56	10	2	2	14	0.04%	0.03%	0.04%	0.11%	4,736	26,272	8,733	13,247	3	8	3	4
472870	49	-	-	17	17	-	-	4	4	-	-	0.10%	0.10%	-	-	4,364	4,364	-	-	4	4
410103	643	289	-	-	289	2	-	-	2	0.32%	-	-	0.32%	23,682	-	-	23,682	144	-	-	144
451924	1190	225	-	-	225	1	-	-	1	0.25%	-	-	0.25%	47,364	-	-	47,364	225	-	-	225

Figure 2.3

2. Scatter plot (Figure 2.4) See referenced document, *Downtime Priorities, Jack-knife Diagrams, and the Business Cycle by Peter F. Knights,* Appendix A.

   The scatter plot is based on a jackknife graph and uses a logarithmic scale to plot MTTR against Event Count. There are limits calculated for both the x and y axis and the plot displays the effect of each CPI on fleet PA. The limits, when displayed on the graph, divide it into 4 quadrants. Three quadrants represent “Chronic and Acute” issues, “Chronic” issues and “Acute” issues. They can be color coded and ranked to provide a clear visual representation of the effect of each CPI issue on fleet uptime. It is preferable to associate ID numbers to each CPI so that the scatter plot is not too busy and becomes difficult to read as the number of units/fleets increase.

   If the project includes multiple fleets, the MTTR is the combined average, and the Event count is the sum of events of all fleets for each specific CPI.

Downtime Stratification Scatter Plot

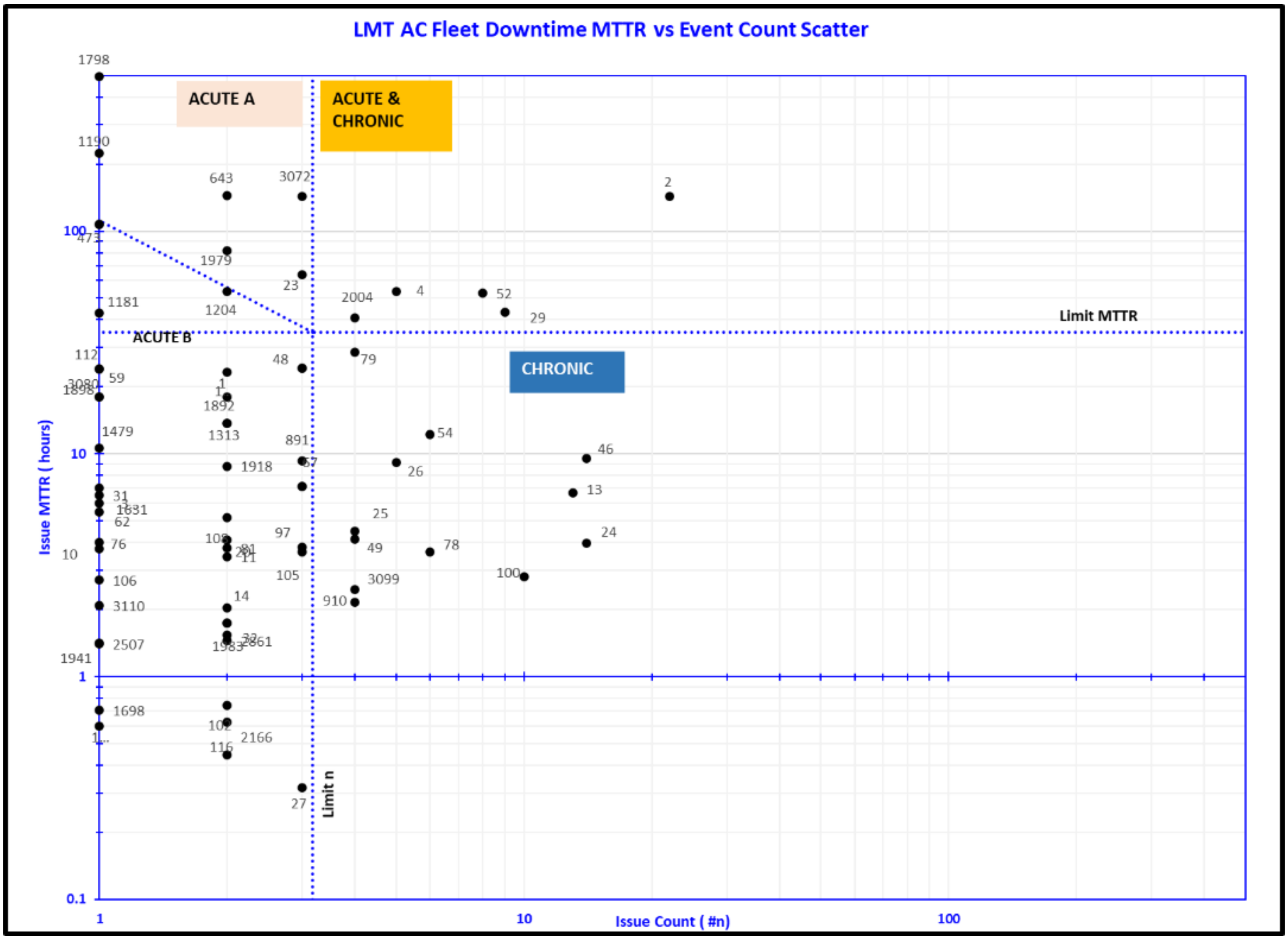

Figure 2.4

3. Top Metrics for each fleet in scope (Figure 2.5). This is a pivot table that is produced from the analyzed data that has associated part numbers and CPI numbers already assigned.

   There are 6 columns of data required from each fleet so that the data can be compiled in the Downtime Stratification Issue list tab.

   If there are multiple fleets, these 6 columns of data should be copied from a separate pivot table for each specific fleet into the Downtime Stratification Workbook to be combined.

   - CPI/NPI
   - PA%
   - MTTR
   - MTBS
   - Downtime Hours
   - Event Count

Fleet 'A' Metrics Summary Sheet

CPI/NPI	PA%	MTTR	MTBS	DT	Count
479365	0.053%	48.9	92194.9	244	5
474605	0.041%	63.8	153675.9	191	3
498557	0.036%	82.0	230527.5	164	2
452506	0.023%	53.4	230556.1	107	2
448166	0.012%	28.0	230581.5	56	2
497956	0.010%	23.4	230586.1	47	2
451107	0.009%	42.9	461176.1	43	1
478157	0.008%	18.0	230591.5	36	2
438190	0.006%	2.8	46119.1	28	10
430935	0.006%	9.3	153730.3	28	3
451937	0.006%	6.5	115298.3	26	4
384757	0.005%	24.0	461195.0	24	1
473472	0.005%	7.1	153732.5	21	3

Figure 2.5

4. The dashboard (Figure 2.6) is an optional tab that summarizes the data collected and analyzed in the other workbook tabs and provides a high-level overview of the project results. This is useful when managing multiple worksites under one project.
   Information recommended in the dashboard include:

   - Dealer
   - Customer
   - Model
   - Number of units
   - Current SMU
   - Target PA
   - Overall Fleet PA
   - Δ PA
   - Δ CPI

Much of this information can be found by searching for the machine serial number in DPP (Digital Performance Plus).

Downtime Stratification Dashboard

Dealer	Customer	Site	Model	Units	Fleet SMU	DPP Month	PA	Avg. R12		Avg. R3		Target	Δ P.A. (R12)	Δ P.A. (R3)	Δ CPI
NA Dealer	NA Customer	Site 1	AC	6	5740	Jul/22	68.2%	77.8%		74.0%		81.9%	-4.1%	-7.9%	5.2%
AU Dealer	AU Customer	Site 2		12	12938	Jul/22	84.0%	83.7%		83.0%		90.0%	-6.3%	-7.0%	3.6%
SA Dealer	SA Customer	Site 3		2	21406	Jul/22	88.1%	81.1%		79.6%		84.0%	-2.9%	-4.4%	6.7%

Figure 2.6

The Stratification Analyst is responsible for:
1. Entering the individual site's KPIs
2. Updating the CPI issue list
3. Keeping the dashboard information current

Additional data that may be useful to your Stratification project include:
1. ICA/PCA information
2. Service Letter Completion Rate
3. Utilization of Availability
4. Customer Type (MARC, DIM, DIWM, DIFM)

3.0 Implementation Steps

The Downtime Stratification project started with a fleet of 794AC trucks operating in Western Canada. The fleet was not meeting PA targets and to determine the root causes, a deep dive was required to find the specific product problems that were contributing to the overall physical availability loss.

The pilot was introduced in July 2021 and a total of 13 electric drive fleets (794/6/8) from all around the world were providing data/metrics within the first month. The data was combined to provide Caterpillar with a global representation of which CPIs were causing the most PA loss. Downtime Stratification can be applied to any machine fleet where downtime is captured and can be implemented immediately. The most critical steps in the process are the first two. All three parties must be invested and aligned and willing to assign team members to do the labor-intensive tasks outlined in steps 3 to 7 below.

Steps Required to Implement this Best Practice

1. Get alignment between Caterpillar, Dealer and Customer
2. Assignment of roles within each organization to handle collection, analysis and sharing of the data
3. Customer collects and reconciles downtime data.
4. Dealer, Customer and Caterpillar analyze minimum 80% of data to assign part numbers causing failure
5. Dealer and Caterpillar assign CPIs to the downtime data
6. Caterpillar team member analyzes data to calculate required metrics
7. Caterpillar Project Analyst compiles fleet metrics and presents results

4.0 Benefits

Caterpillar Benefits

- Direct correlation between CPI Rankings and impact to Customer experience (machine physical availability)
- CPI resource allocation
- Visibility to chronically unreported issues (hose that fails 100 times)
- Line of sight to management to effective ICA definition and implementation

Dealer Benefits

- Sharing field corrective action with other participating dealers while awaiting a PCA
- Identifying when ICA or PCA solutions have not been implemented
- Identifying areas to improve fleet downtime such as for rework, PMs, and waiting on parts/labor.
- Identifying high frequency, low duration failures that cause as much downtime as the one-off, long duration events
- Support prioritization of backlog execution vs impact to physical availability

Customer Benefits

- Increased fleet reliability and availability
- Increased efficiency of labor resources
- Lower cost per hour and cost per ton
- Improved MTTR and MTBS

5.0 Resources Required

Customer engagement is a vital step in Downtime Stratification. Providing detailed stoppage records to the dealer is a crucial step in the process. The dealer must have a representative that can compile and sort the raw data from the customer should it be required so that it is in a useable format. The time spent on this task depends on factors such as the complexity of the raw data and the fleet size. On average, it is expected to take 1 hour per week. Another task performed by the dealer representative includes collaborating with the Cat team and the customer, preferably in a 1 hour long weekly meeting, to analyze line by line, 80% of the DT and assign part numbers/missing details and CPI numbers. The Caterpillar SPM or Tech Rep is then required to spend an additional half hour per week (up to 2 hours per month) organizing the stratified data and calculating the metrics. A Caterpillar Stratification Analyst compiles the data. Both Microsoft Excel and Power BI can be used as a tool by the Stratification Analyst for computing the data and/or presenting the results. On average, each of the Customer, the dealer and Caterpillar, invests 8 hours per month to Downtime Stratification. Table 5.0 summarizes the required resources.

Resources Required

<table>
<tr><th>Personnel Resource</th><th>Task</th><th>Estimated Duration (hours/month/fleet)</th></tr>
<tr><td>Customer - Maintenance & Operation contact</td><td>Reconcile D/T data captured throught site management system</td><td>4</td></tr>
<tr><td>Dealer - Reliability Engineer / Fleet Analyst</td><td>D/T data to Excel format</td><td>4</td></tr>
<tr><td>Customer - Maintenance & Operation contact</td><td rowspan="3">Assign CPI and/or part numbers to 80% of the D/T data</td><td rowspan="3">4</td></tr>
<tr><td>Dealer - Reliability Engineer / Fleet Analyst</td></tr>
<tr><td>Caterpillar - SPM or Tech Rep</td></tr>
<tr><td>Caterpillar - SPM or Tech Rep</td><td>Analyze D/T data and calculate Top Tier Metrics</td><td>2</td></tr>
<tr><td>Caterpillar - Stratification Analyst</td><td>Compile metrics and other pertinent information into the Downtime Stratification Workbook</td><td>2</td></tr>
</table>

Table 5.0

6.0 Supporting Attachments / References

Appendix A: Downtime Priorities, Jack-knife Diagrams, and the Business Cycle by Peter F. Knights
Appendix B: Downtime Stratification Workbook

7.0 Related Best Practices

N/A

8.0 Acknowledgements

This Best Practice was written by:

Iesha Courtney
Technical Representative
Courtney_iesha_a@cat.com
1-780-714-8506

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